Northeastern University

EECE 3392: Electronic Materials

Memristor: The Fourth Passive Component

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We attest that this report contains work derived from our Team members wherein every attempt has been made to avoid plagiarism and to assign appropriate credit to sources of information.

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1. Background and Conception of the Memristor

In the domain of electronic components, there exist three passive components responsible for calculating the relationship of voltage, current, charge and flux in a circuit: resistors, inductors and capacitors. The resistor can be described as a component whose physical properties relate the current through a branch as modeled by Ohm's Law: $dV = Rdi$. A capacitor links the voltage across its terminals to the charge retained by its electrodes, modeled as $dV = Cdq$, and the inductor links the current through itself to the rate if change of magnetic flux as $d\Phi = Ldi$. These three components have long been familiar to engineers and physicists. Their properties, parasitics, manufacturing, and how they interact with one another have been studied extensively and understood for well over a century.

It was not until recently that a fourth fundamental passive element connecting the rate of change of charge to the rate of change of flux was proposed: the memristor. The concept of the memristor emerged in 1971, theorized by Leon Chua, a professor at the University of California, Berkeley. Chua proposed that the fourth passive element related charge to flux by the equation: $d\Phi = Mdq$ (Domaradzki et al., 2020). This property was conceptualized to produce a component that could change its resistance based on current that has once passed through the device. It wasn't until 2008 that the first physical memristor was developed by a team at Hewlett-Packard Laboratories, led by R. Stanley Williams. This invention marked a significant milestone, not just in electronic engineering, but also in the industry of computing and artificial intelligence.

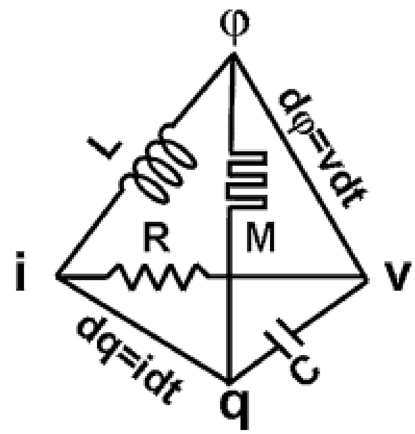


Fig. 1 Diagram displaying the relation of the four fundamental passive components

1.1 Memristor Basics and Operation

Fundamentally, a memristor represents a new type of passive two-terminal electronic device, defined by its ability to "remember" the amount of charge that has flowed through it over time. Unlike the property of resistance, which obeys Ohm's Law, capacitance, which describes the storage of charge in an electric field, and inductance, which describes the storage of energy in a magnetic field, memristors possess a distinct property called "memristance". This property implies that the electrical resistance of a memristor is dependent on the amount of charge that has passed through it previously. This creates a device that is capable of remembering a previous state of itself, without the need for power. In other words, memristors can be considered a form of non-voltable information storage.

To comprehend the working principle of a memristor, an understanding of the materials used to create these devices and their nanoscale structures is required. Most memristors are fabricated using metal oxides; The most successful candidate has typically been titanium dioxide. These metal oxides exhibit a phenomenon known as "ionic drift" when an electric field is applied. In metal oxides, oxygen vacancies can occur due to imperfections in the crystal lattice structure of the material. When an electric field is applied across the terminals of a memristor, it induces a flow of charge through the device resulting in a current along with the movement of these positively charged oxygen vacancies in response to the field gradient (Wang et al., 2015). This migration of ions results in changes to the material's electrical resistance, deriving the memristive property. The operation of a memristor can be explained by its different states: high resistance and low resistance. When no voltage is applied across the terminals of a memristor, it remains in its high resistance state. Upon the application of a voltage, an electric field is generated, inducing the migration of oxygen vacancies. The vacancies present in the crystal will accumulate or dissipate, depending on the polarity of the applied voltage. The resistance of the memristor decreases or increases, respectively, transitioning it to the low resistance (ON) or high resistance (OFF) state. The migrated ions within the material retain their state after the removal of the applied voltage, hence the non-volatile nature of the memristor.

1.2 Basic Mathematical Model

A more in depth method of understanding the memristor is to mathematically describe its behavior in the current-voltage plane. As stated prior, a memristor will have a low on resistance and high off resistance. The method of constructing a memristor involves bonding two metal oxides, one of which has been ionically doped to reduce its resistance, one that remains with its intrinsic resistance. These two joined doped and undoped oxides comprise the two separate resistive on and off states of the memristor respectively. This is physically depicted by figure 2 as pictured on the left (Wang et al., 2015). The simplest mathematical model of a memristors IV behavior is based off of Ohm’s Law, with the R variable replaced with the equation determining a memristor’s resistance for a

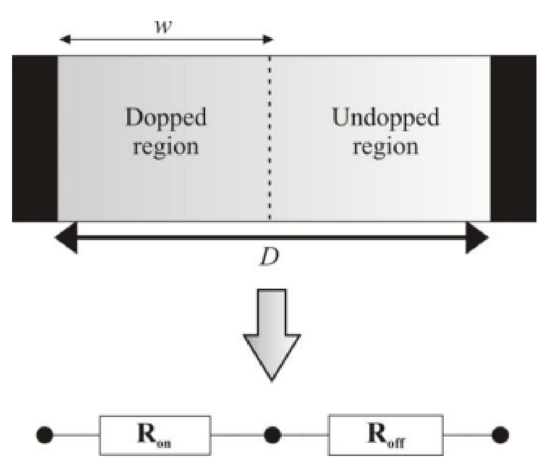


Fig. 2 Diagram depicting the physical structure of a memristor, and variables involved in its dimensions

given internal state variable w(t). The internal state variable is the value of the length of the doped region of the memristor. Due to physical limitations, the oxygen vacancies, which control the internal state of the memristor, are unable to move as freely at the edges of the structure, so their mobility is diminished. For the purpose of understanding the memristor, the following equation has been simplified under the assumption that the vacancies can always move freely.

$$(1)\ \frac{dw(t)}{dt} = \frac{\mu_v R_{on}}{D}\, i(t)$$

$$(2)\ V(t) = \left(R_{on}\frac{w(t)}{D} + R_{off}\left(1 - \frac{w(t)}{D}\right)\right) i(t)$$

Equation 1 describes the relation between the internal state variable and the electrical charge mobility, μ_v. This is tied into the equation for the total resistance of the memristor shown in equation two, describing the resistance as a function of the sum of the total lengths Ron and Roff comprise of the total structure length, D

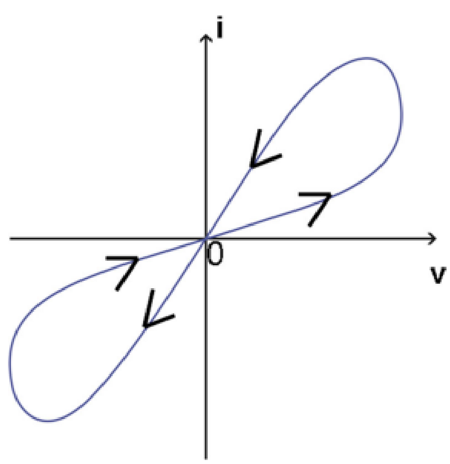


Fig 3. I-V curve of memristor as described by equation 2. Displays the pinched hysteresis loop created by its behavior.

(Domaradzki et al., 2020). Expressed in the current-voltage plane, equation 2 forms a pinched hysteresis loop, visualizing the theoretical IV curve of a memristor, as depicted in fig. 3.

1.3 Broad Scope of Memristors and Greatest Potential

The memristor's ability to "remember" its resistive state is pivotal for various applications, especially in the field of non-volatile memory and neuromorphic computing. In memory applications, memristors can be utilized to create high-density, low-power storage devices that retain data even when the power is turned off. They show great promise in their capability to mimic synaptic plasticity, the ability of synapses in the brain to strengthen or weaken over time. This makes memristors an excellent potential candidate for neuromorphic computing, facilitating the development of artificial neural networks with the possibility to learn and adapt to input stimuli (Ivanov et al., 2022). This paper will delve further into the physical and chemical workings of the memristor, its use in neuromorphic computing, and the market trend outlooks for the devices in the near future.

2.1 Materials Used in Memristors (Electrodes)

One of the critical aspects influencing the performance and functionality of memristors is the choice of materials used in their fabrication. The memristor mainly has a metal-insulator-metal (MIM) sandwich structure that includes two layers of electrodes and a middle resistive layer formed by a semiconductor or insulator. The properties and resistance of the memristor is closely related to the component materials of the device.

The electrodes in memristors serve not only to conduct electric current but also to participate in the resistive switching (RS) reaction. Various materials are commonly used for memristor electrodes, including conventional metals (like gold and platinum), noble metals (like silver and palladium), alloys (like Copper-Tellurium), carbon-based materials (like graphene), nitrides (like TiN and TaN), and transparent conductive oxides (such as indium tin oxide (ITO)).

From here, the electrode materials can be categorized into three types based on their role in the RS behavior. First, inert metals such as gold (Au) and platinum (Pt) mainly function as current conducting media and have minimal effect on the RS behavior. These materials usually have high conductance, stability, and durability so as to properly function over long periods of time or over many uses. Inert metals are preferred in situations where stability, reliability, and predictability of the resistive switching behavior are essential.

Inert metals are the most commonly used materials in memristors for storage and logic circuits. In these applications, precise control over the resistance state of the memristor and long term stability are required. These memristors also offer non-volatile memory storage capabilities, allowing for data retention when no power is being supplied. These materials offer the most reliability over many cycles, leading to their abundance of use in the industry as they are seen being integrated into semiconductor devices as well.

Second, electrochemically active metals such as copper (Cu) and silver (Ag) are responsible for the formation of conductive filaments through electrochemical dissolution and deposition, particularly in cation migration-based memristors. These metals act as catalysts, creating pathways for current to flow that influences the resistive state of the memristor, making the overall device more volatile. Thus, these kinds of materials are more commonly found in neuromorphic computing applications, a form of computing that aims to emulate the functionality of the human brain, as they more closely mimic the synaptic plasticity of a biological neural network.

These electrochemically active metals allow for cation migration-based devices to exist. These devices contain migrating, positively charged ions (cations) in the material, leading to the formation and dissolution of conductive pathways, which in turn changes the resistive state. Electrochemically active metals participate in redox reactions when they undergo a resistive switch state, essential for the migration of cations. These migrating cations allow for dynamic and reversible changes in the resistive state, mimicking the synaptic plasticity and learning capability of the brain (neuromorphic computing). Memristors with dynamic and non-linear resistive switching behaviors thus can react and adapt to changing input stimulus, allowing for the ability to "learn" from prior events and to be able to modify behavior over time, similar to biological synapses. Other applications for electrochemically active materials in memristors include pattern recognition and adaptive control systems. This can be seen in autonomous driving vehicles where they must constantly be scanning a changing environment for obstacles and correctly respond in an efficient and safe manner.

Finally, novel electrodes like indium tin oxide (ITO), F-doped tin oxide (FTO), and graphene, for example, are used in flexible and transparent memristors. Flexibility is desirable for applications such as flexible displays, wearable electronics, and flexible sensors that are required to bend and conform to irregular surfaces while retaining full functionality. Flexibility also increases durability as it will not be damaged if any force is applied to the sensor that would cause it to bend. Transparency is vital as it is used in touchscreens, solar cells, and smart windows where maintaining visibility through the device is important.

More specifically, the applications of novel electrodes are seen in biomedical devices

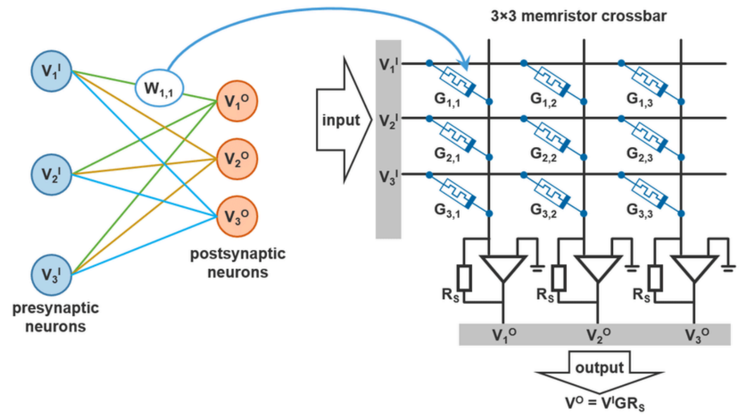


Fig. 4 Model of neural network using memristors as memory to adapt to different input stimuli

such as biosensors, implantable devices, and wearable health monitors. These devices must conform to biological surfaces so as to be non-invasive. They must also maintain transparency to not obstruct the view underlying tissues or muscles during imaging or monitoring. Another application for this material is found in neuromorphic computing, a highly elastic system that would require the flexibility and transparency aspects given by novel electrodes to properly emulate its complexity.

Overall, the choice of electrode material significantly impacts the performance and behavior of memristors, influencing factors such as stability, flexibility, transparency, and CMOS compatibility.

2.2 Materials Used in Memristors (Resistive Switching)

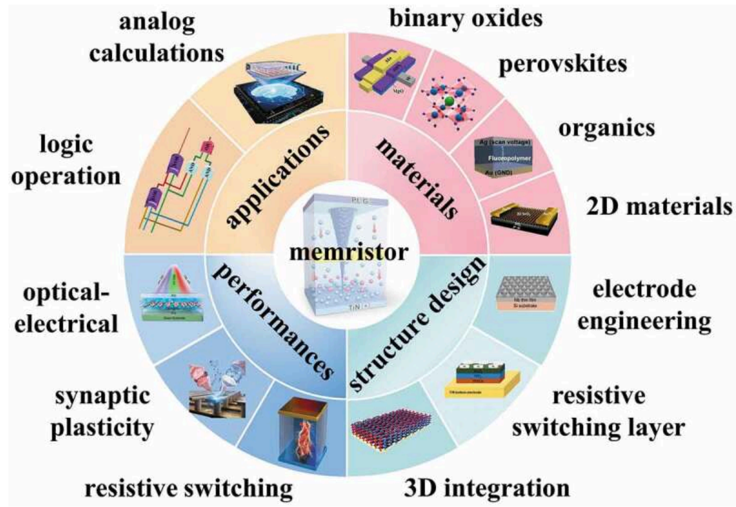


Fig. 5 Four aspects of memristors broken down into core categories

Next, the resistive switching material is the core of memristor technology. Different materials have different resistive characteristics and thus affect how the memristor behaves. The main materials used can be classified into two separate categories: inorganic and organic materials. Inorganic materials such as binary oxides, perovskite, and 2D materials, for example, are typically associated with more stable, robust, and faster resistive switching behavior. Organic materials are associated with a higher flexibility, low cost, and ease of production.

Binary oxides represent a significant category of inorganic materials in memristor technology, characterized by their simple composition, high stability, low cost, and compatibility with traditional semiconductor manufacturing processes like CMOS. Among the many available materials, HfO (hafnium oxide) and TaO (tantalum oxide) have emerged as particularly promising due to their sub-nanosecond operation speed and exceptional endurance, surpassing 10^{10} cycles. Despite these favorable characteristics, conventional binary oxide memories often fall flat due to the necessity for high power consumption or low uniformity. Researchers thus developed innovative solutions to these challenges with the creation of bismuth-doped tin oxide (Bi: SnO_2) memristors, which demonstrated significantly reduced operating currents and enhanced resistance values compared to conventional devices.

With the rapid development of portable and wearable devices as well as efforts to optimize memristor performance for neuromorphic computing applications, the development of flexible memristors is paramount. Scientists have already created a resistance random access memory (RRAM) with bilayer TiO_2/HfO_2 structures which exhibits 500 times of mechanical bending without degradation in performance. Scientists have also created complex three-layer structures like $HfO_2/BiFeO_3$(BFO)/HfO_2 memristors which demonstrate improved resistive switching performance and multi-level storage capabilities, enabling highly accurate pattern recognition.

Organic materials are presented as one of the more promising answers to the demand for flexible electronic systems due to their low cost, solution processability, flexibility and ability to be created in large-scale preparation.

Significant strides have been made in trying to enhance organic memristors' performance and endurance. In this realm, Liu et al. developed an $Ag/2DP_{BTA+PDA}/ITO$ memristor, demonstrating good switching performance and high reliability, with a drawback of its endurance being limited to a mere 200 cycles. To address this limitation, Xu et al. introduced a robust diffusion memristor based on a copolymer of trifluorochloroethylene and vinylidene fluoride (FK-800), enhancing the switching endurance to be over 10^6 cycles.

Despite improvements in endurance, manufacturing yield remains a challenge for organic memristors. Zhang et al. achieved a record yield of 90% for polymer memristors by implementing a 2D conjugation strategy, resulting in low power consumption and miniaturization. Additionally, Park et al. developed a solution-processed flexible organic memristor array with self-selectivity, achieving high recognition accuracy in neural networks.

Novel organic materials like deoxyribonucleic acid (DNA) and egg proteins have garnered attention for their biodegradability, biocompatibility, and low cost. Xu et al. developed a DNA-bridged memristor with silver nanoparticles, exhibiting low operating voltage, minimal power consumption, and high switching speed. Basic logic calculations were successfully demonstrated, showcasing the potential of these organic materials in information processing systems.

3. Practical Applications & Advantages of Memristors

As seen, memristors have loads of different advantages that have the potential to revolutionize the electronics industry. Due to its passive nature, the memristor naturally sees a low power loss while still being compatible with both digital and analog environments. It will also hold its memory state when the power is off, providing even more power efficiency. Furthermore, their non-volatile nature proves to be extremely useful when it comes to developing computing systems, as memristors can inherently act as memory. This allows for easy integration with modern CMOS logic and parallel processing with arrays. All of this could potentially allow the continuation of Moore's Law: providing us with faster, easier to produce, and more power/size efficient chips.

3.1 Integration With CMOS Inverter

In order to show the potential of computing-based memristive capabilities, they would have to be implemented into a CMOS-level design. This is what Vishwakarma, et al. sought to accomplish in 2023 with their memristive-CMOS inverter. The inverter is one of the most used logic gates in digital computing, with NOT, NAND, and NOR logic blocks being heavily used. The successful implementation of memristive technologies in an inverter would substantially lower the size of computing systems while at the same time making them faster and wasting less power.

A basic model of this memristive-CMOS inverter can be seen below in figure 7, with a traditional CMOS inverter being shown in figure 6.

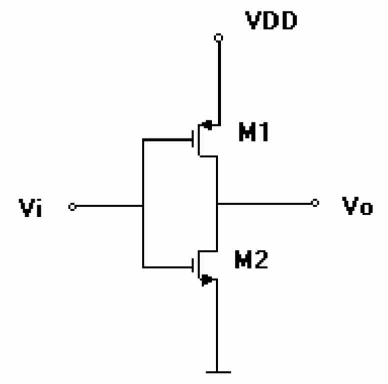


Fig. 6 Traditional CMOS inverter

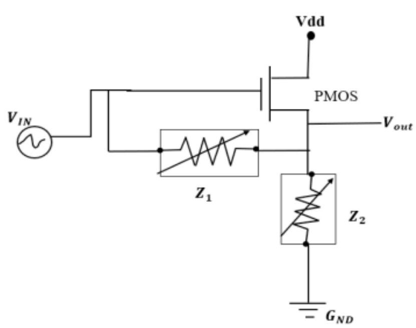


Fig. 7 Hybrid Memristor-CMOS Inverter

As seen in the above figures, the hybrid CMOS inverter is effectively the normal CMOS inverter, except the N-channel MOSFET (M2) was replaced by a memristor-divider circuit that provides feedback to the P-channel MOSFET (M1). This memristor sub-circuit allows for the saving of a previous state in conjunction with assisting the PMOS with processing new information.

When Vin = 0, both the PMOS and the memristors will be in an ON state: the PMOS will allow VDD to sink through to Vo, inverting the output. When Vin = 1, the PMOS and Z1 will turn off, but Z2 will stay on. As the on-state resistance is lower than the off-state resistance, Z2 will sink VDD to ground instead of Vo, successfully inverting the input. Simulation results can be seen in figure 8.

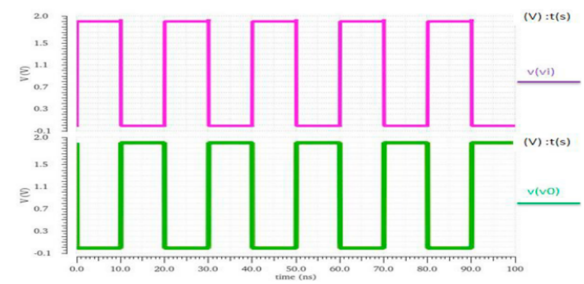


Fig. 8 Hybrid CMOS Inverter Input vs. Output

Overall, these researchers showed that it is theoretically possible to integrate memristors with existing CMOS architecture designs with relative ease. They showed that using 130nm CMOS technology and 1.9V supply voltage, this design would only consume 35.67uW of power and would switch with a 9.235ps delay. However, this all depended on a circuit simulation that used memristor models.

3.2 Other CMOS Integration

While Vishwakarma proposed a relatively simple model, it goes to show how easy hybrid integration was while keeping things to a fairly basic level. Other, larger-scale, attempts have been made to combine CMOS and memristive technologies; notably in the form of Cai et al., who created a fully reprogrammable integrated memristor-CMOS chip in 180nm CMOS technology. This more system-level approach allowed for easy integration of a memristor crossbar array with CMOS computing. The chip consisted of ADCs and DACs alongside the CMOS circuits and memristive crossbar array.

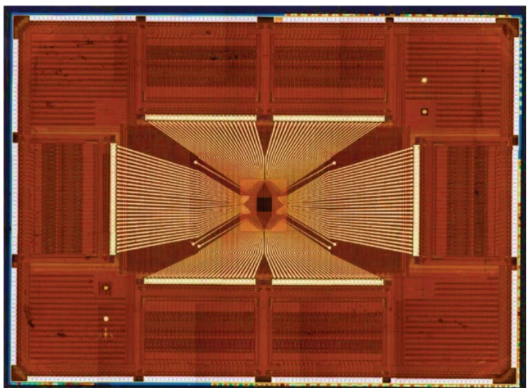


Fig. 9 Microscopic Image of Memristive Crossbar Integrated with CMOS Circuitry

This setup allowed for the parallel processing of information and memory within the memristive CMOS chip. By setting up this parallel processing architecture, the efficiency of the chip is greatly increased. As a result, the power consumption of this chip is in the tens of milli-watts range with this number expected to decrease as these hybrid systems become more and more efficient.

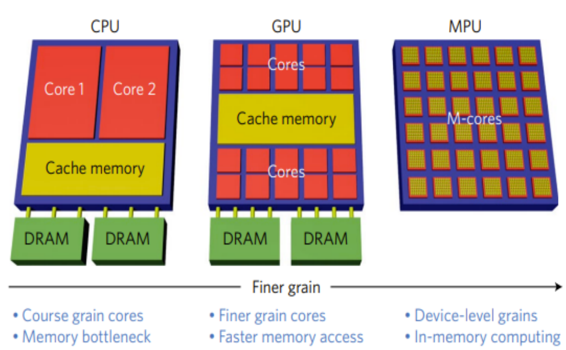


Fig. 10 Progression of Processor Architecture, From CPU to MPU

The main takeaway from these hybrid memristive-CMOS circuits is that they allow processors to store their memory and processing circuitry internally; meaning that general computing speed is greatly enhanced. Figure 10 shows the progressions from CPU to GPU to “MPU” (Memristive Processing Unit). It is also important to note how much smaller the MPU can be compared to the alternatives, given its combination of processing and memory into one circuit.

3.3 Neuromorphic Circuits and Artificial Neural Networks

Over the past decade, more and more research has been poured into mimicking and studying brain activity through different means. A neuromorphic circuit, in theory, would be able to replicate the human brain in terms of both processing and memory capabilities. Memristors have a clear role here as in order to create an artificial brain, you need to mimic the processing functionality of the brain. As it turns out, the brain works by having a large network of synapses connect various neurons. These synapses web out to potentially hundreds of thousands of neurons as the brain processes new information in parallel. Figure 11 shows the difference between a typical computing system and a “brain” computing system.

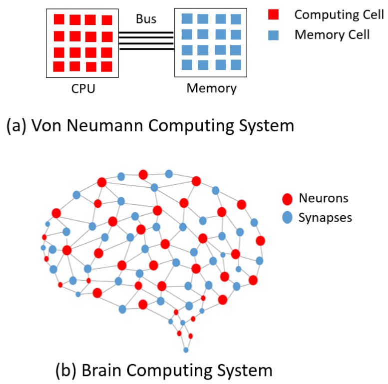


Fig. 11 Traditional vs. Brain Computing System

To create such a brain computing system, one can connect a cross bar memristive array to a CMOS architecture with a similar setup to what was shown in section 3.2. Huang et al. used this exact system, managing to produce exceptional results with extremely high speeds and efficiency. The memristors have also shown easy cross-compatibility with CMOS circuitry in both function and mass-production. However, time is still needed to polish everyday memristors and it is still unclear when, or if, these hybrid circuits will become adopted by chipmakers.

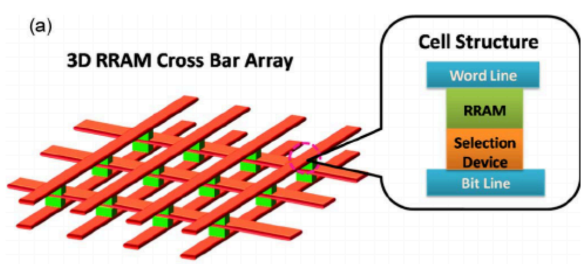
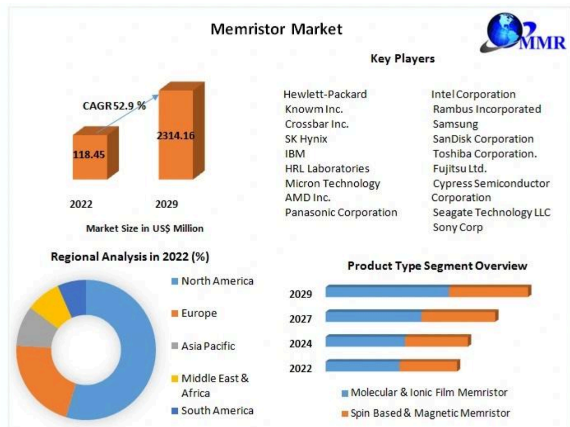


Fig. 12 Basic Crossbar Array

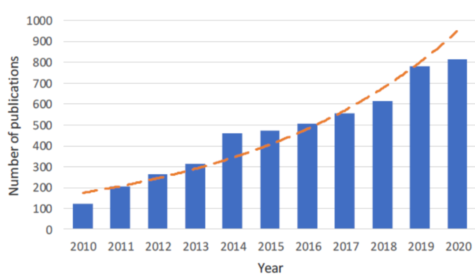
4. Market Analysis and Future Prospects

Current market prospects for memristors are very promising. The estimated market size for memristors in 2023 was $185 million USD, but it is expected to grow to over $2 Billion USD by 2029. This estimated jump is due to the relative recency of commercially viable memristive devices, as the first ever programmable memristor computer was made in 2019 by a group of researchers at the University of Michigan Ann Arbor.



Currently, the key players in the memristor market are Hewlett-Packard, Samsung and Toshiba Corporation. The majority of memristor analysis is conducted in North America, though this analysis is likely to shift more to Asia as the Chinese government begins to research memristive devices more. Currently, molecular and ionic film memristors make up the majority of commercially available devices and are expected to continue doing so as we move further into the future.

Assuming that memristive devices become cheaper and easier to mass produce, the two largest applications for memristors will be in artificial neural networks and data storage. Artificial neural networks have been experiencing a recent boom in popularity as humanity begins to better understand biological and man-made neurology, the latter field referred to as neuromorphic computing. Memristors have the ability to remember which electric state it was in after being turned off, have similar function to a biological synapse, meaning that they can be chained together like neurons in the brain. Currently, this technology hasn't been researched much as it was just discovered in August 2023, but more and more papers are being published each year.



Memristor device publications by year

Currently the market size for neural network technologies is around $200 Billion USD, but neural networks are beginning to hit the limitations of current technology. Most neural networks today are currently based on the Von Neumann architecture meaning that all processing is sequential and that computation and memory are separated. While this works great for generating information from previous data, when asked to create something new, current models fall short. This is where memristors come in, as due to their ability to store data and allow current to flow through, they enable parallel processing with collocated processing and memory, allowing for more advanced neuromorphic networks to emerge in the future.

The other major limiting factor in the advancement of AI is data storage - also the other major application for memristive devices. Given that the market size for global data storage is ~$186.75 Billion USD, power consumption is a critical cost for any large-scale operation to consider. Data storage and retrieval centers accounted for 3% of global electricity use in 2023 - 40% more than the United Kingdom - and are expected to only increase as neural networks begin creating more and more information.

This is where the memristors come into play - researchers in Israel and China found that memristive circuits used "between 1/800th and 1/5000 the energy of a conventional computer processor" when performing basic operations. Their non-volatile nature allows for data retention without continuous power supply, allowing for significant reductions in energy consumption and heat dissipation.

Memristive devices are also easily scalable and compatible with existing silicon-based technologies, making them a viable candidate for next-generation storage architectures. For example, HP has already prototypes a crossbar latch memristor memory that can fit 100 gigabits in a square centimeter and have already proposed scalable 3d designs that can hold up to 1 petabit per cubic centimeter.

4.1 Current Limitations

While memristor technology is very promising, there are still many limitations to surpass before the technology can be fully utilized. Despite many published scientific papers, the complete model describing resistive switching phenomenon is still missing, and thus the ideal memristor has not yet been found. Also, as mentioned earlier this technology only became commercially viable 5 years ago in 2019, and those designs are not easily scalable.

The field of memristor-based neuromorphic computing is similarly still in its infancy. Since neural networks with parallel processing on a large scale are few and far between we are still limited to simple memory and binary logic demonstration. This, combined with the recency of memristor technologies has stalled investment into memristor-based
the field of memristors looks bright.

neuromorphic computing. Also, building more complex memristor-based neural networks requires more complex neuron designs and better control and understanding of memristor dynamics, though this is expected to improve as we move further into the future.

5. Concluding remarks

Memristors have the potential to shape future electronics, and it's not hard to see why. While there are many limitations to surpass before we achieve commercially viable memristor devices, with advancements in modern technologies we're getting closer and closer to finding the ideal memristor model. The current global memristor market is expected to grow by 30% each year for the next 7 years, and with applications in analog circuitry, neuromorphic computing and data storage

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